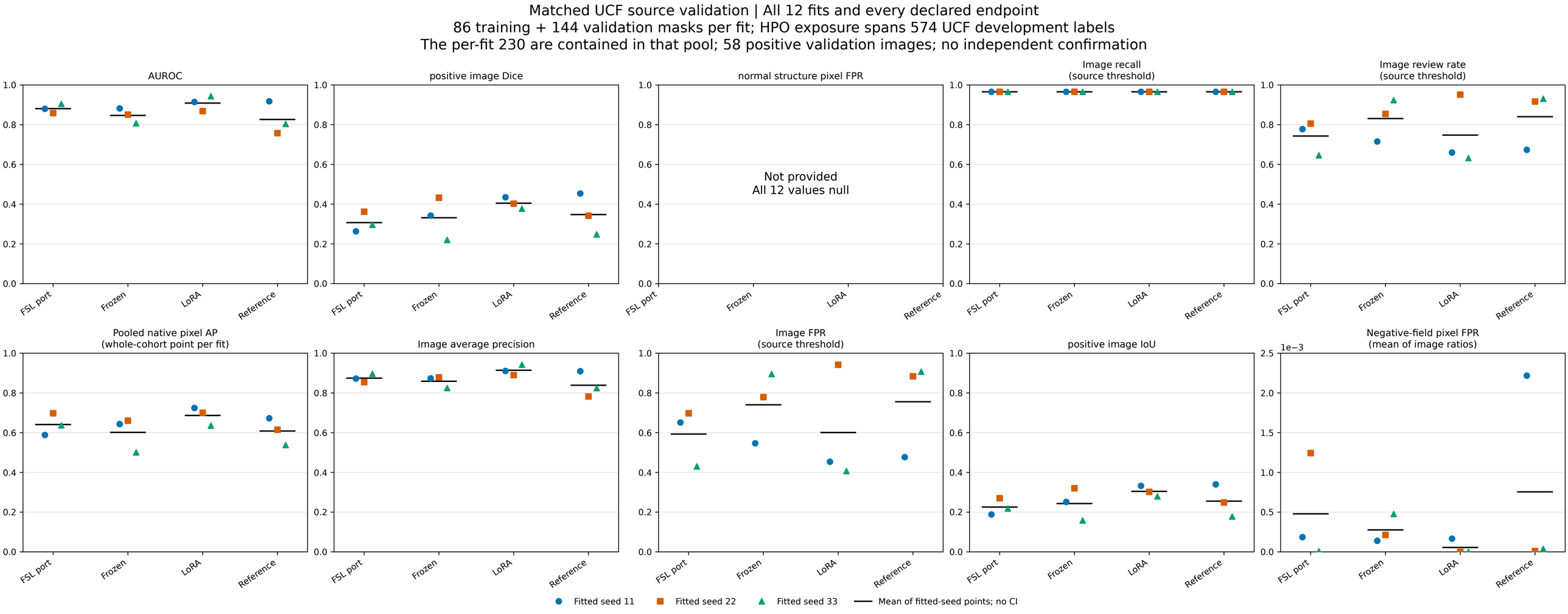
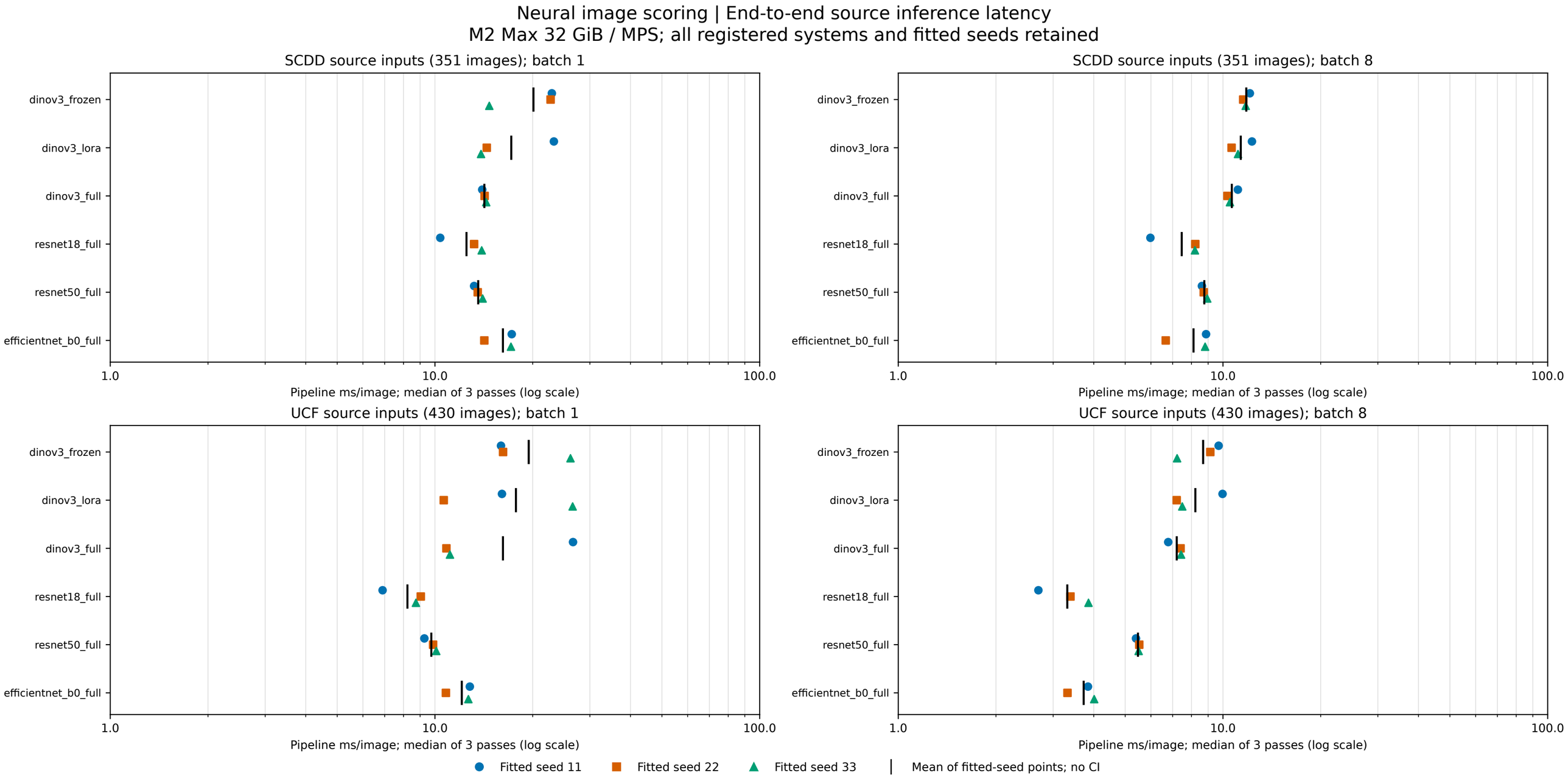
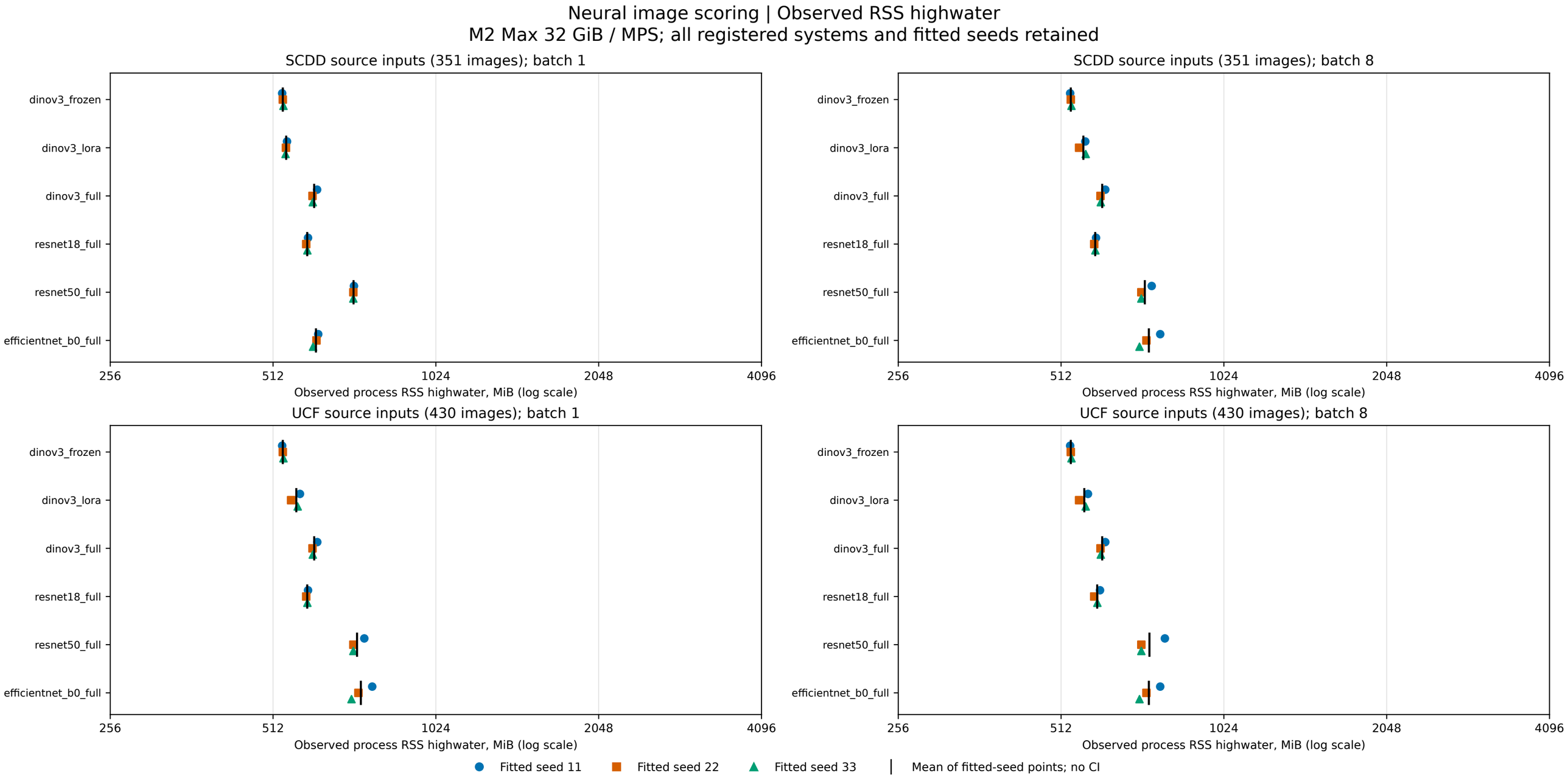
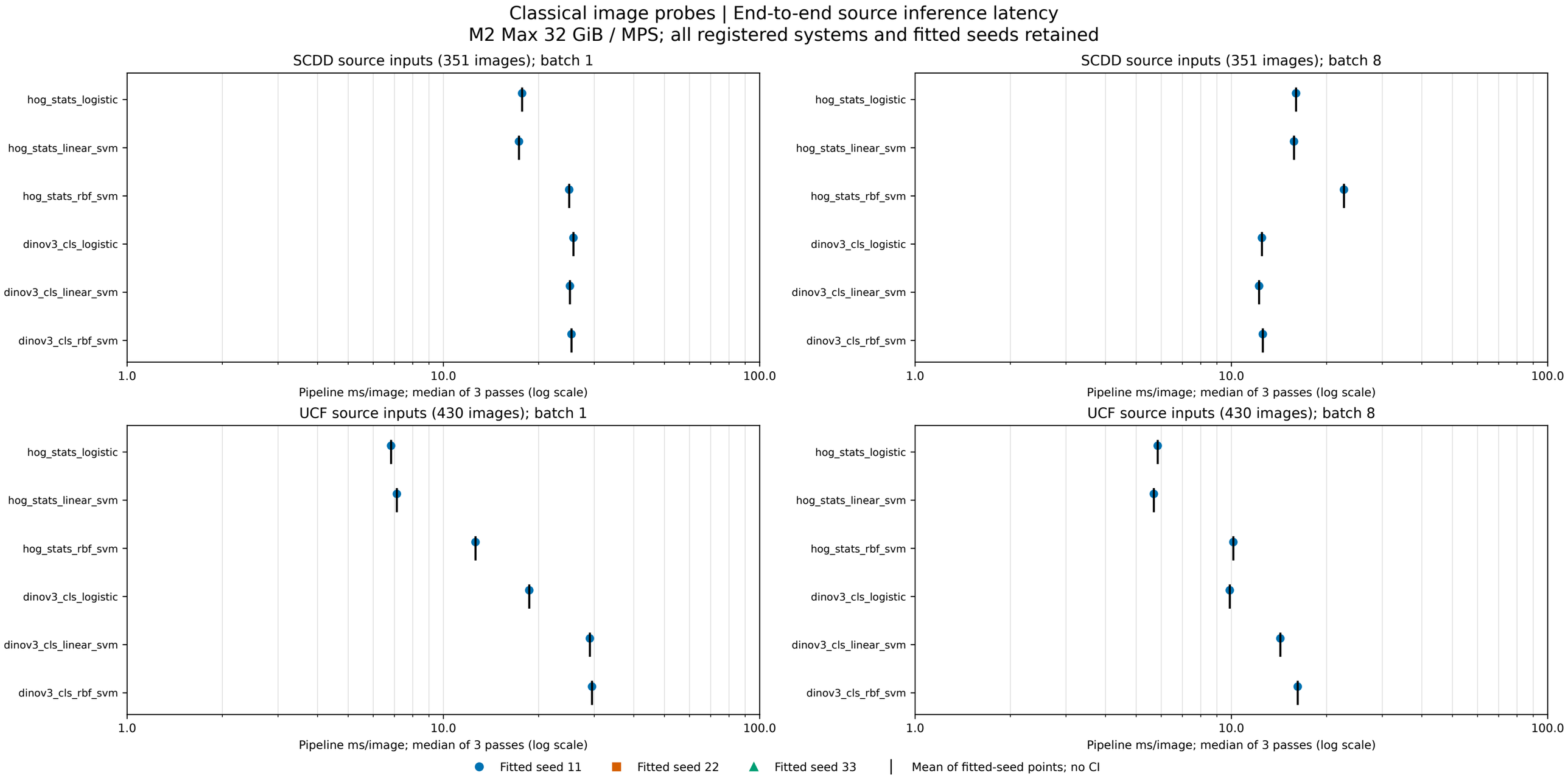
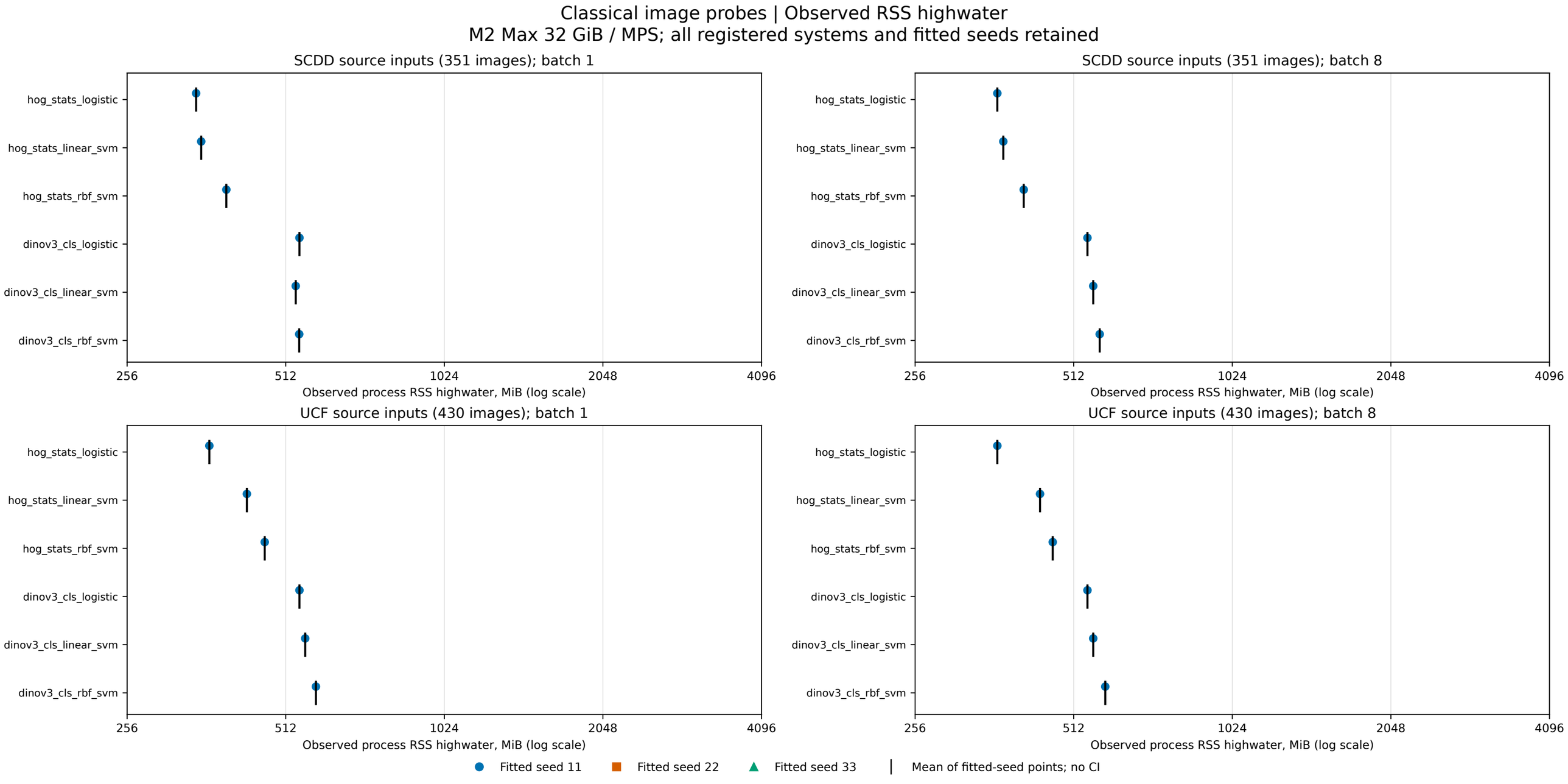
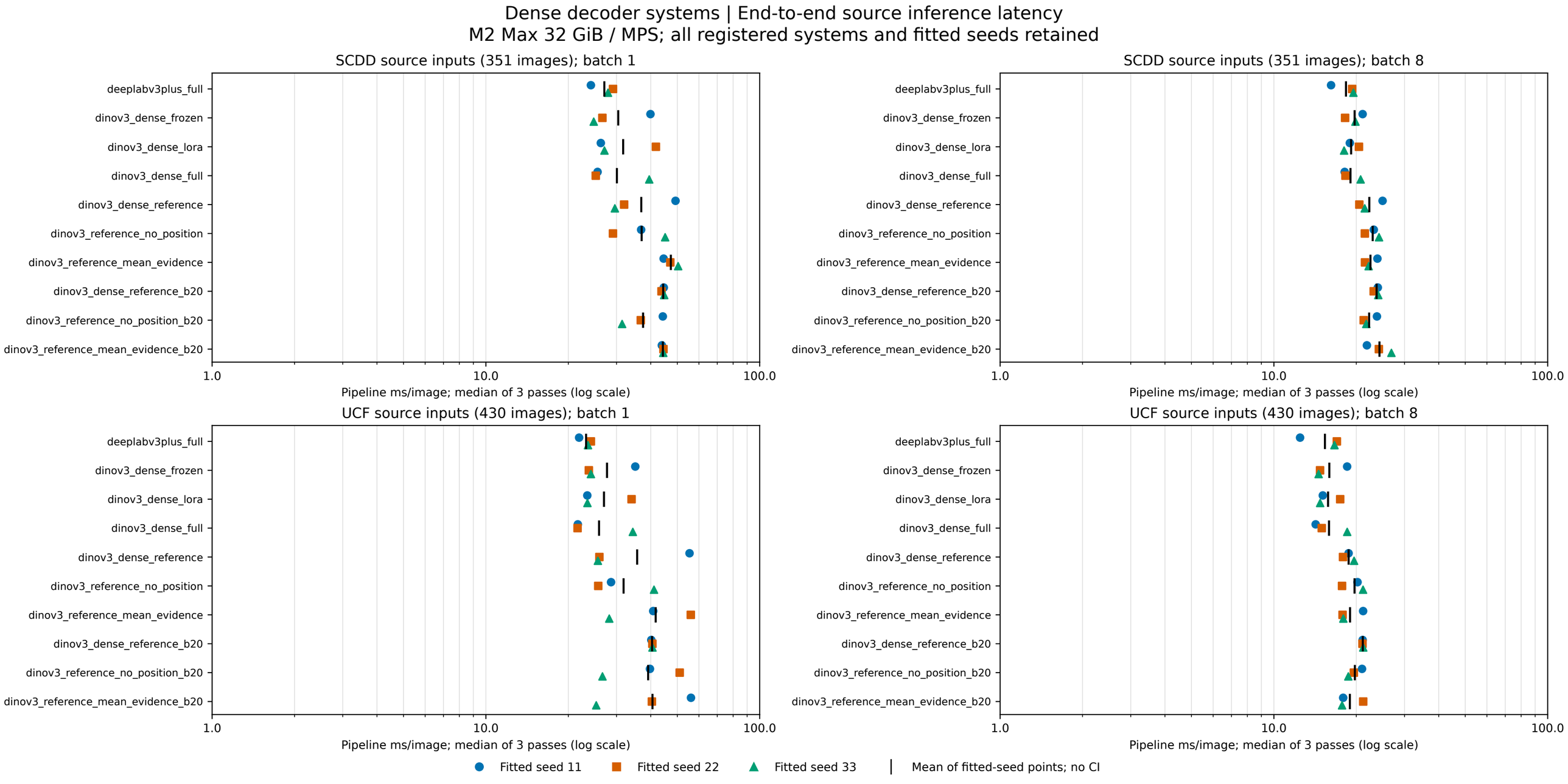

Matched UCF source validation | All 12 fits and every declared endpoint
86 training + 144 validation masks per fit; HPO exposure spans 574 UCF development labels
The per-fit 230 are contained in that pool; 58 positive validation images; no independent confirmation
AUROC
positive image Dice
normal structure pixel FPR
Not provided
All 12 values null
Image recall
(source threshold)
Image review rate
(source threshold)
Pooled native pixel AP
(whole-cohort point per fit)
Image average precision
Image FPR
(source threshold)
positive image IoU
Negative-field pixel FPR
(mean of image ratios)
1e−3
FSL port
Frozen
LoRA
Reference
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI

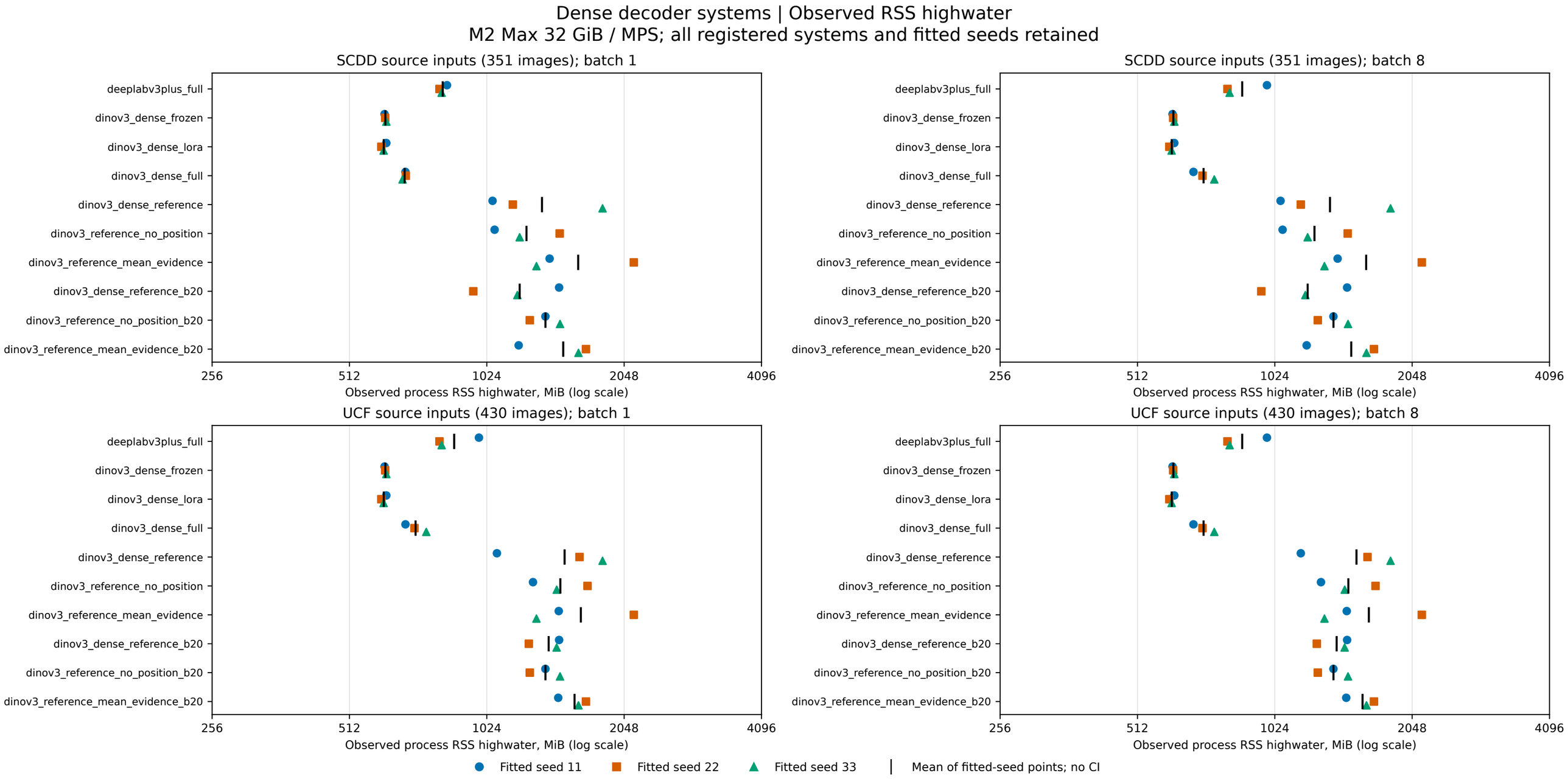

Neural image scoring | End-to-end source inference latency
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
dinov3_frozen
dinov3_lora
dinov3_full
resnet18_full
resnet50_full
efficientnet_b0_full
1.0
10.0
100.0
Pipeline ms/image; median of 3 passes (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


Neural image scoring | Observed RSS highwater
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
dinov3_frozen
dinov3_lora
dinov3_full
resnet18_full
resnet50_full
efficientnet_b0_full
256
512
1024
2048
4096
Observed process RSS highwater, MiB (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI

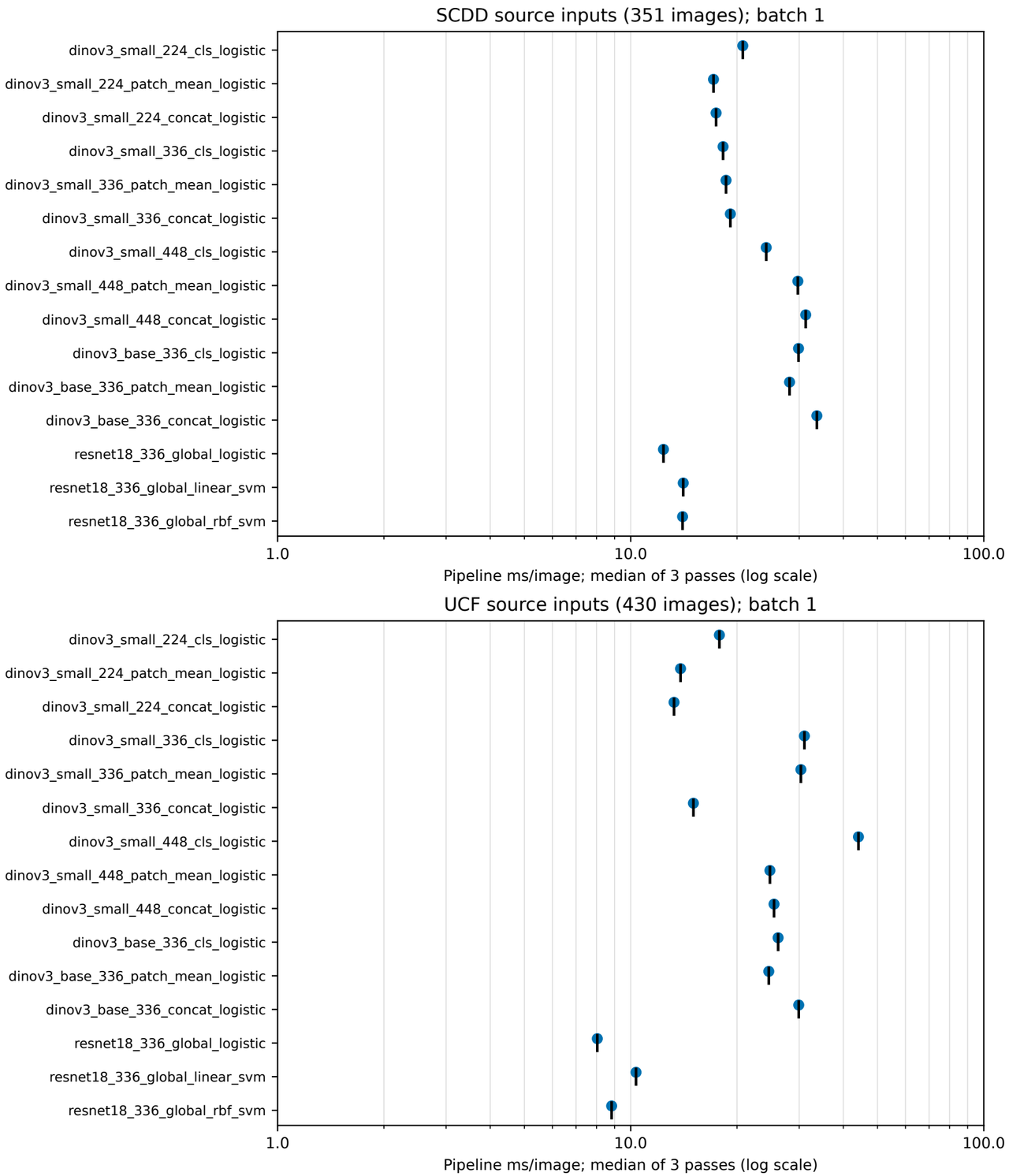

Classical image probes | End-to-end source inference latency
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
hog_stats_logistic
hog_stats_linear_svm
hog_stats_rbf_svm
dinov3_cls_logistic
dinov3_cls_linear_svm
dinov3_cls_rbf_svm
1.0
10.0
100.0
Pipeline ms/image; median of 3 passes (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI

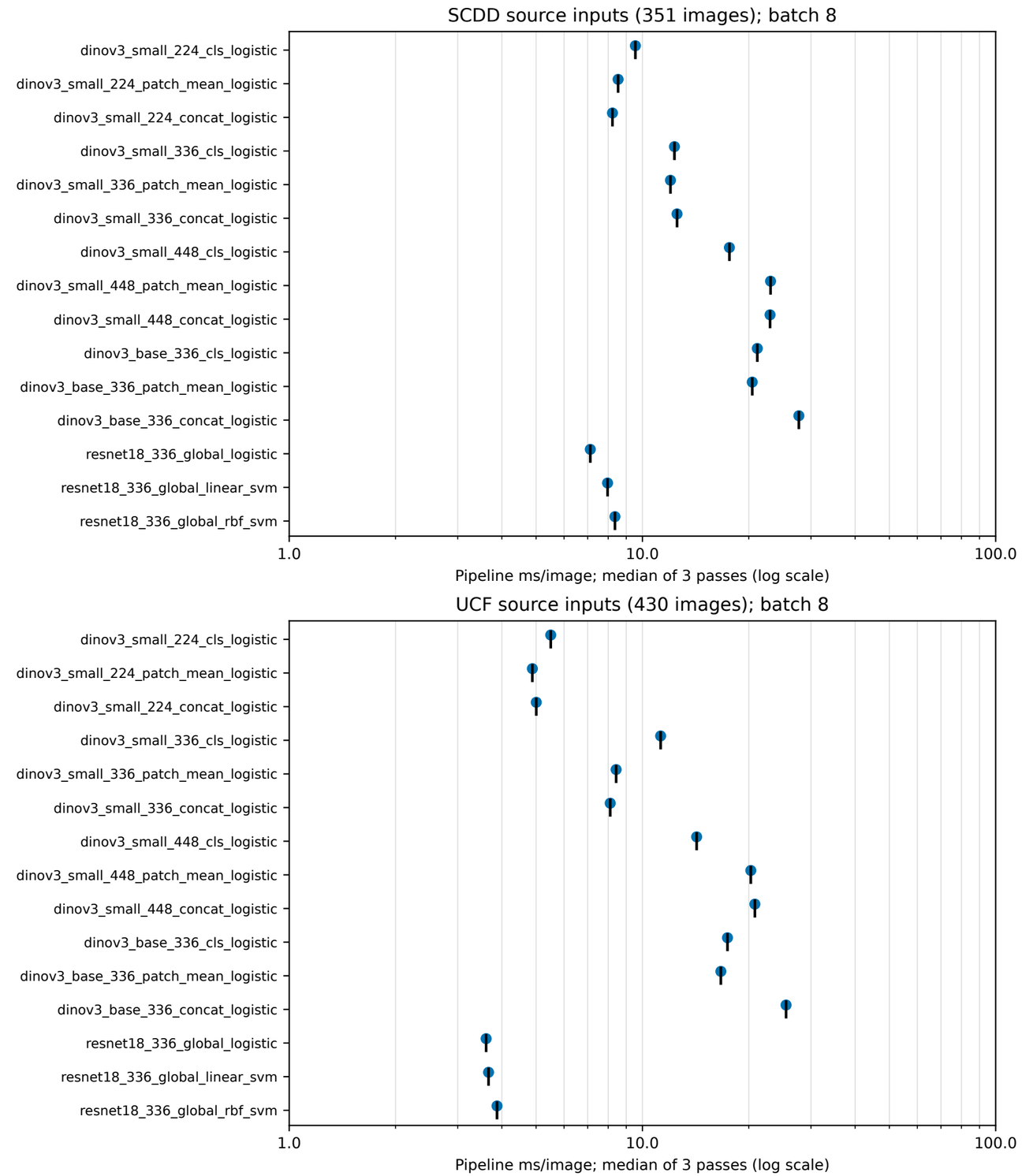

Classical image probes | Observed RSS highwater
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
hog_stats_logistic
hog_stats_linear_svm
hog_stats_rbf_svm
dinov3_cls_logistic
dinov3_cls_linear_svm
dinov3_cls_rbf_svm
256
512
1024
2048
4096
Observed process RSS highwater, MiB (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


Dense decoder systems | End-to-end source inference latency
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
deeplabv3plus_full
dinov3_dense_frozen
dinov3_dense_lora
dinov3_dense_full
dinov3_dense_reference
dinov3_reference_no_position
dinov3_reference_mean_evidence
dinov3_dense_reference_b20
dinov3_reference_no_position_b20
dinov3_reference_mean_evidence_b20
1.0
10.0
100.0
Pipeline ms/image; median of 3 passes (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


Dense decoder systems | Observed RSS highwater
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
deeplabv3plus_full
dinov3_dense_frozen
dinov3_dense_lora
dinov3_dense_full
dinov3_dense_reference
dinov3_reference_no_position
dinov3_reference_mean_evidence
dinov3_dense_reference_b20
dinov3_reference_no_position_b20
dinov3_reference_mean_evidence_b20
256
512
1024
2048
4096
Observed process RSS highwater, MiB (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI

Resolution, pooling and backbone probes | End-to-end source inference latency
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained

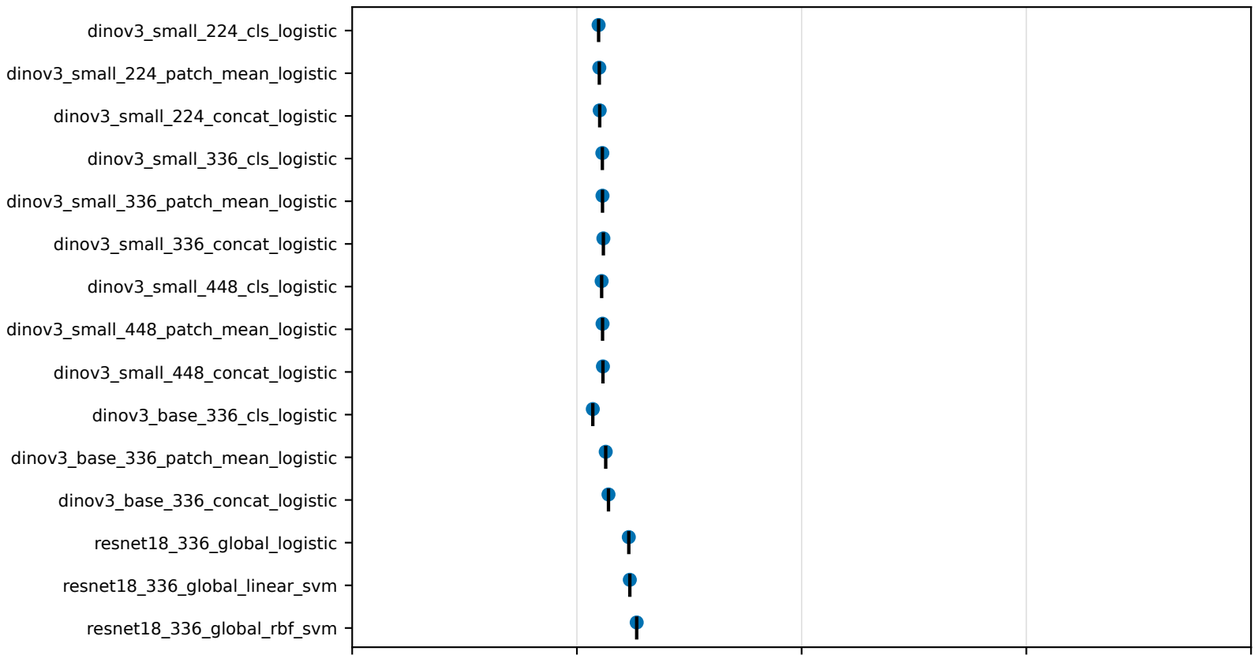




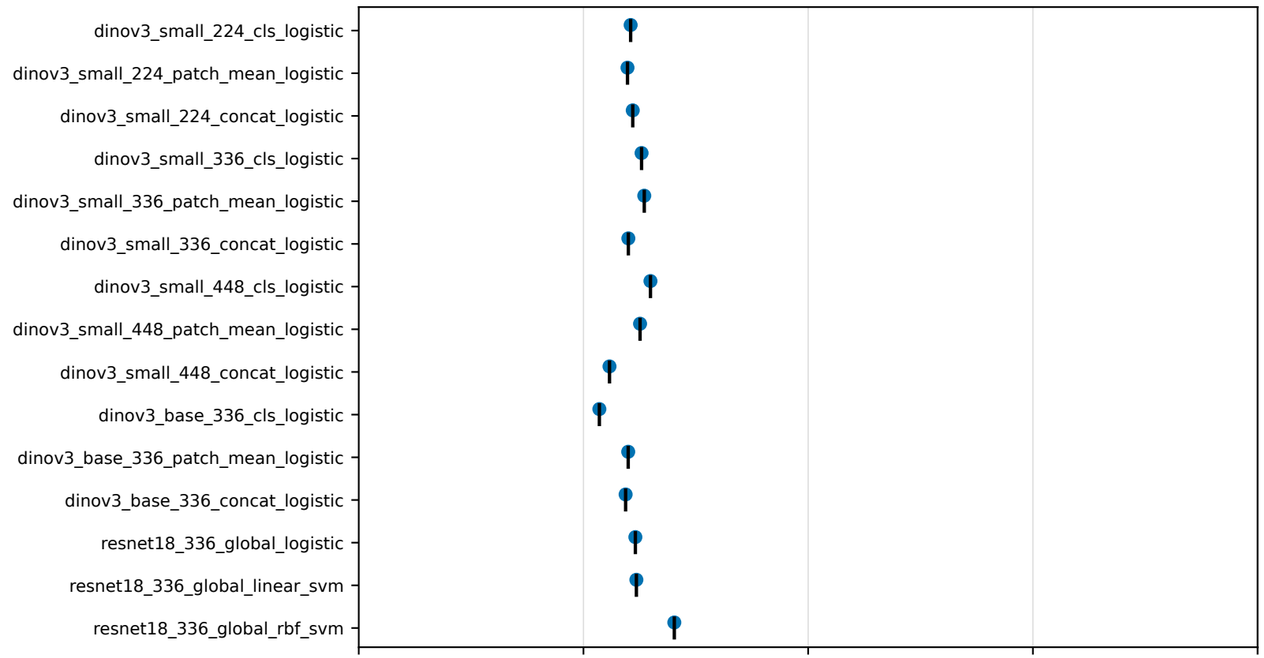
Fitted seed 11 Fitted seed 22 Fitted seed 33 | Mean of fitted-seed points; no CI

Resolution, pooling and backbone probes | Observed RSS highwater
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained

SCDD source inputs (351 images); batch 1

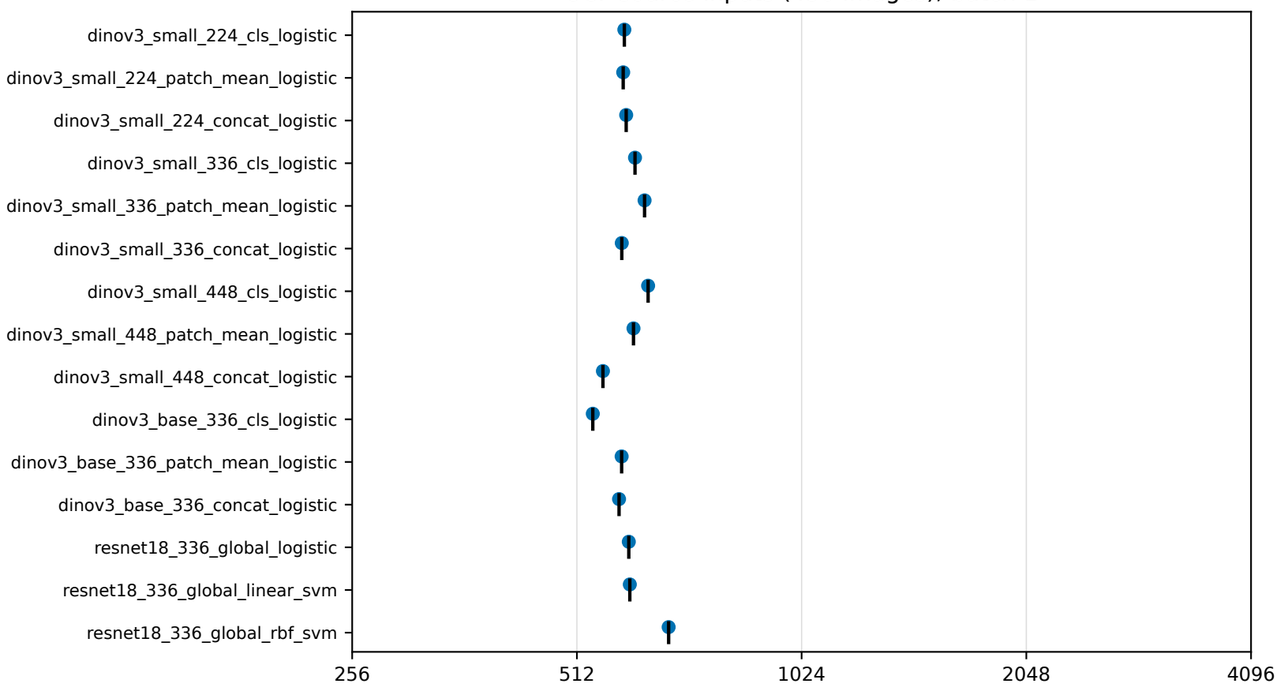


SCDD source inputs (351 images); batch 8



Observed process RSS highwater, MiB (log scale)

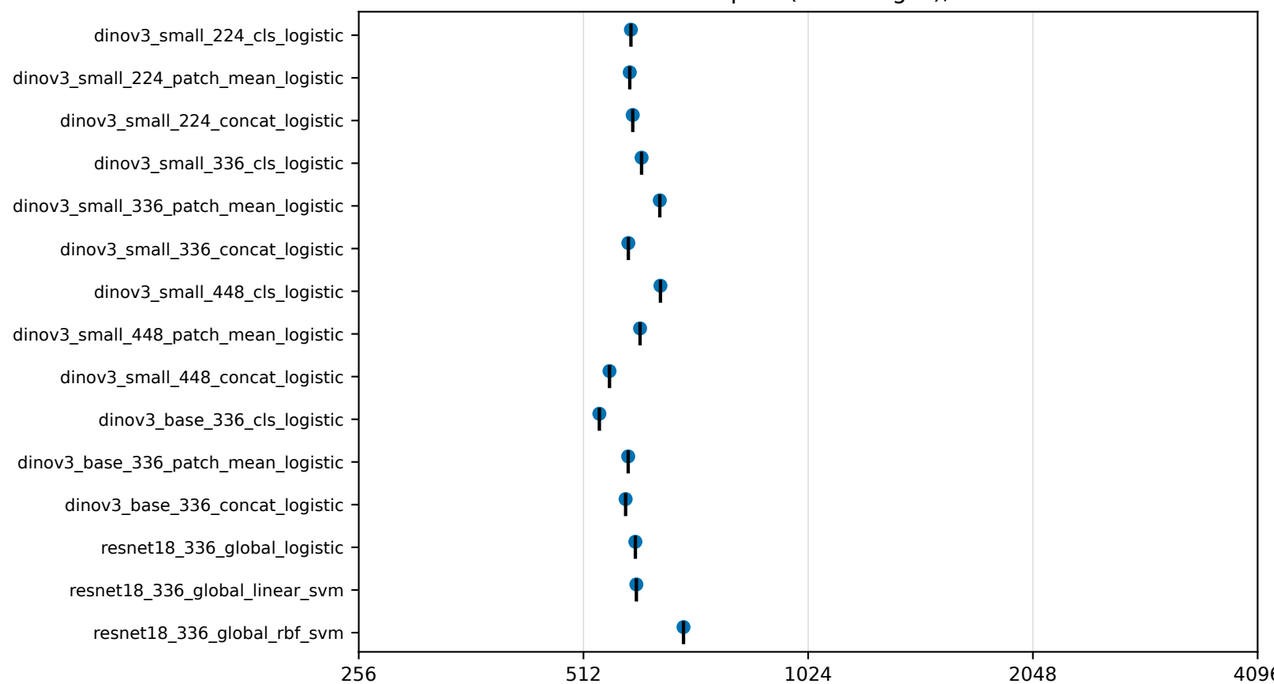
UCF source inputs (430 images); batch 1



Observed process RSS highwater, MiB (log scale)

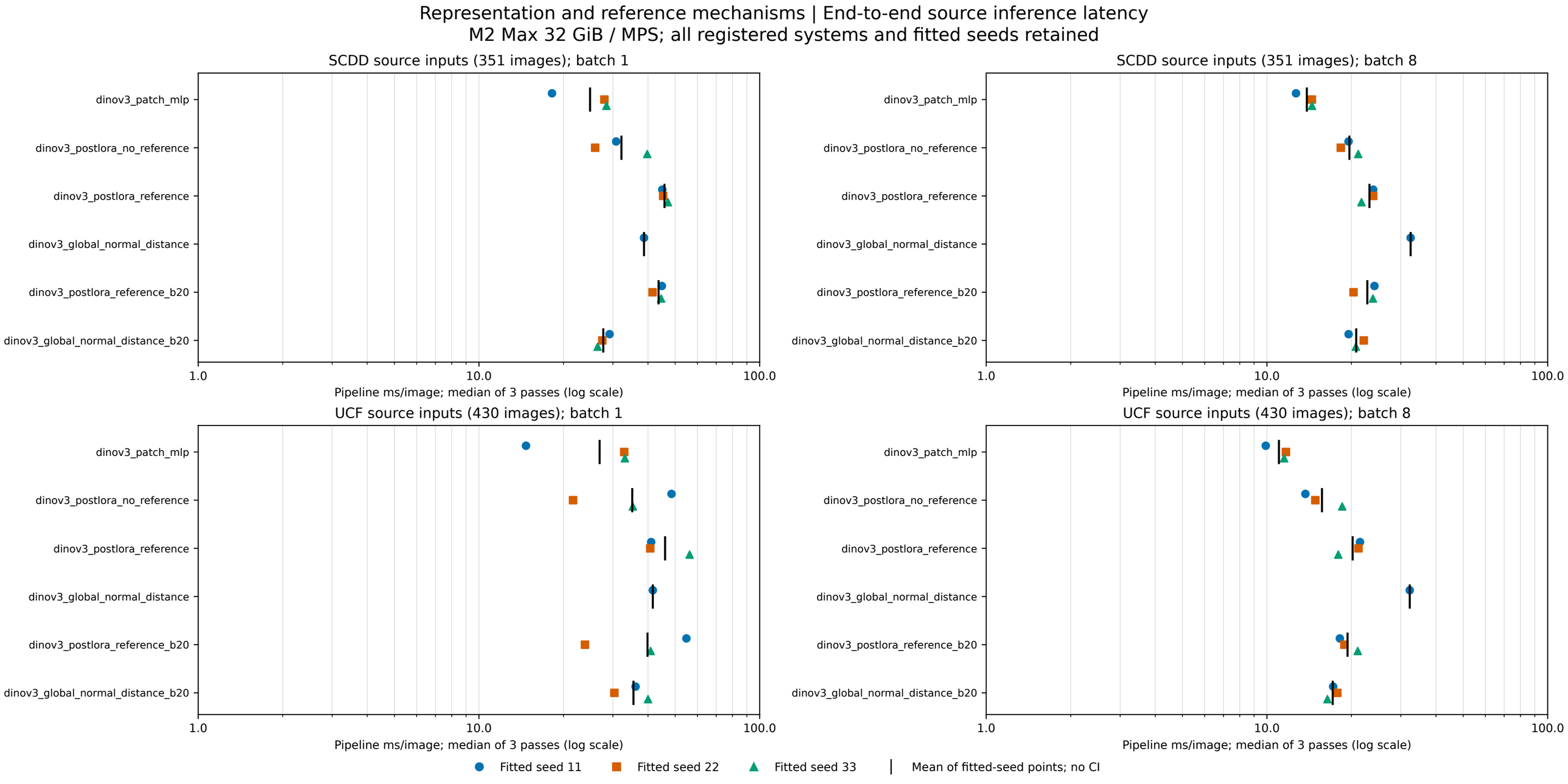
Observed process RSS highwater, MiB (log scale)

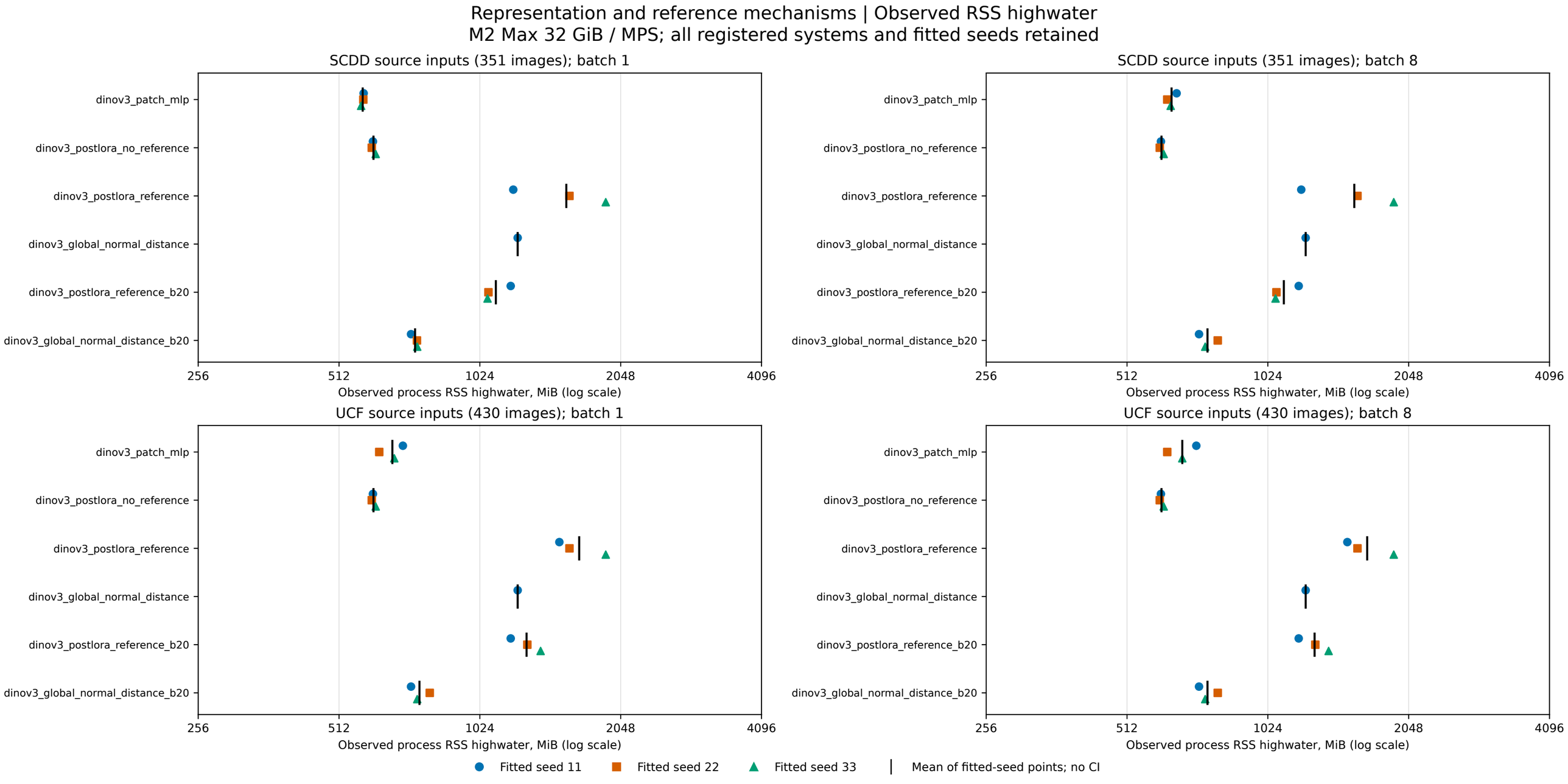
UCF source inputs (430 images); batch 8

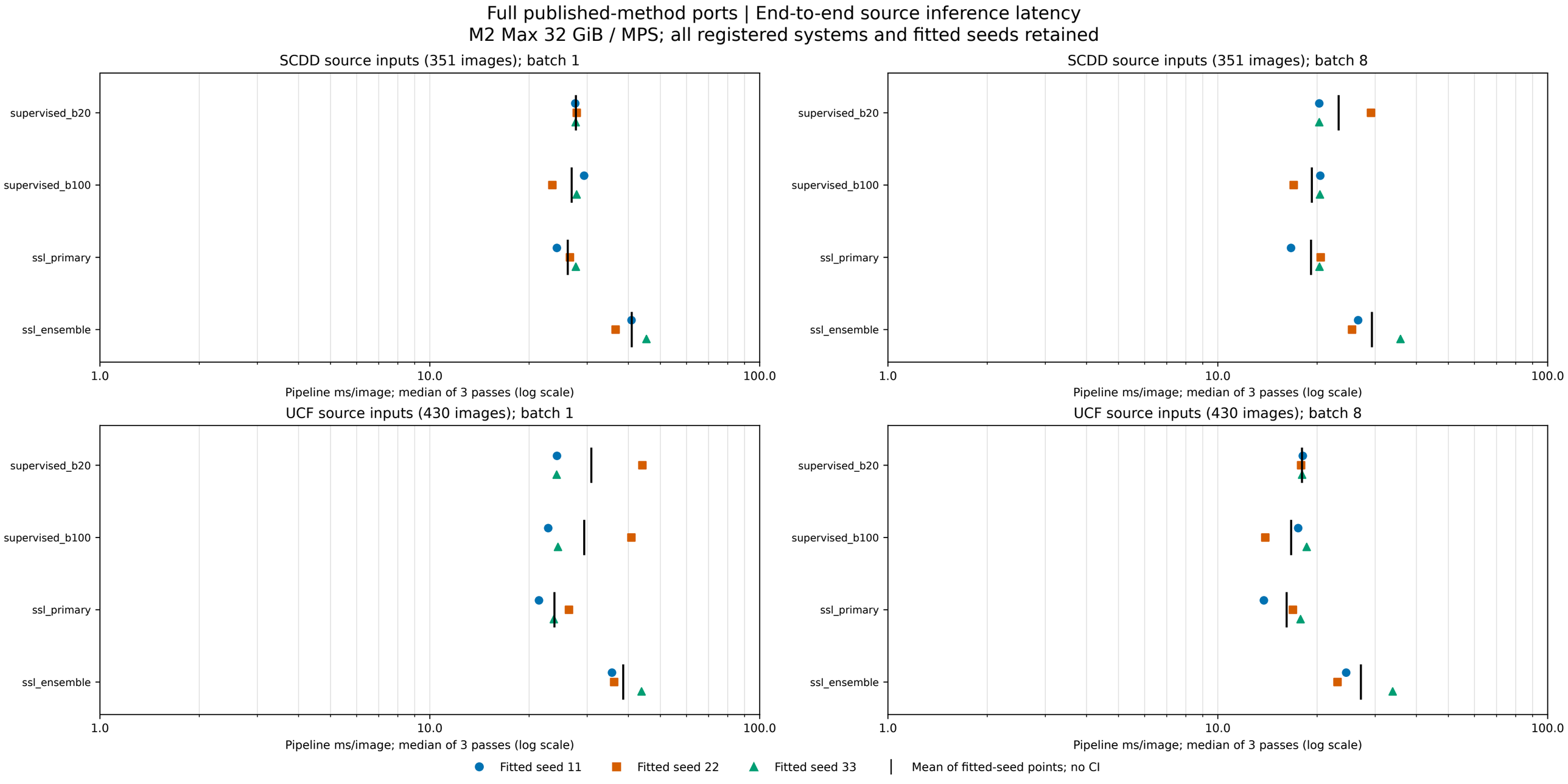


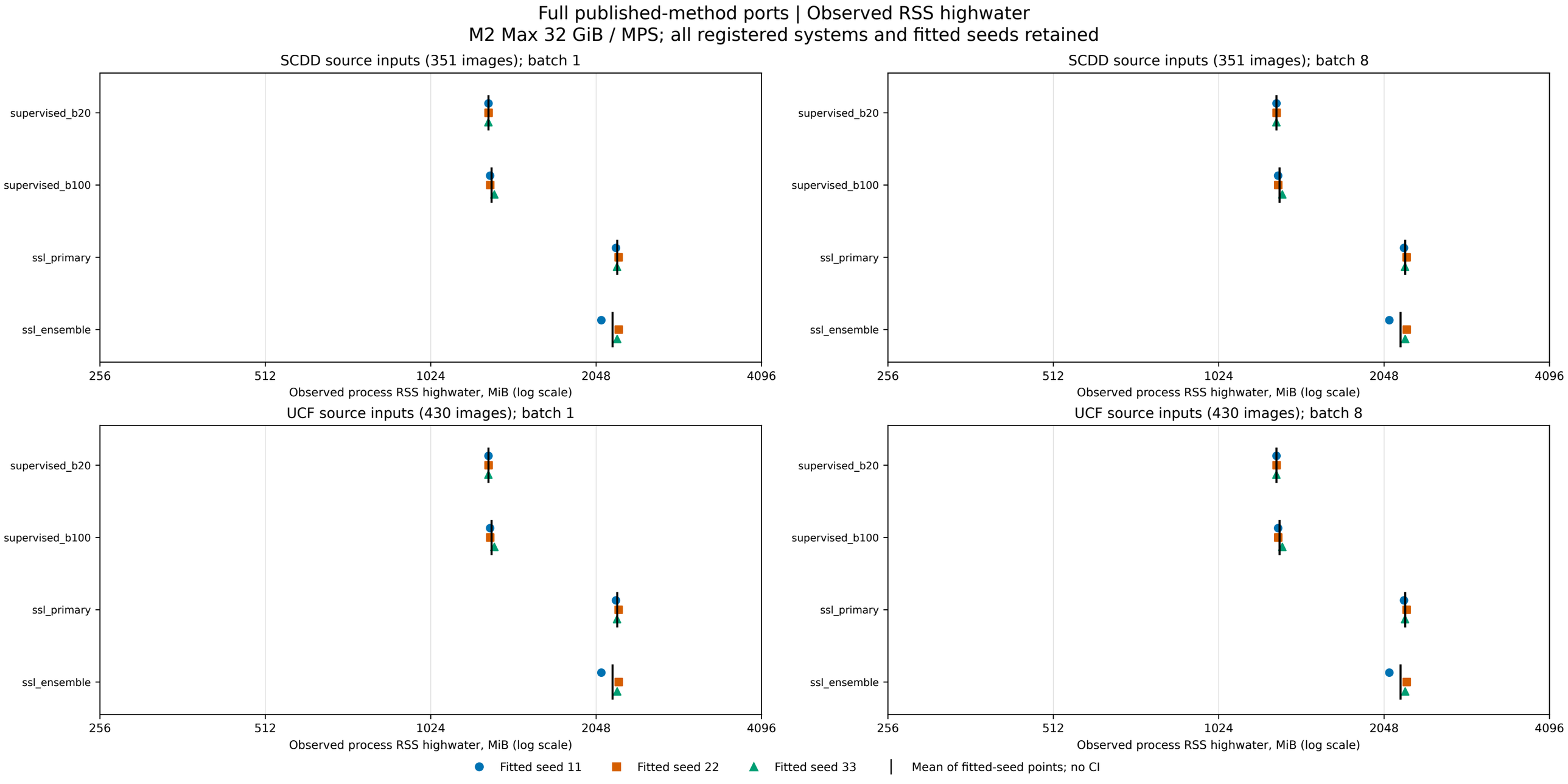
Observed process RSS highwater, MiB (log scale)

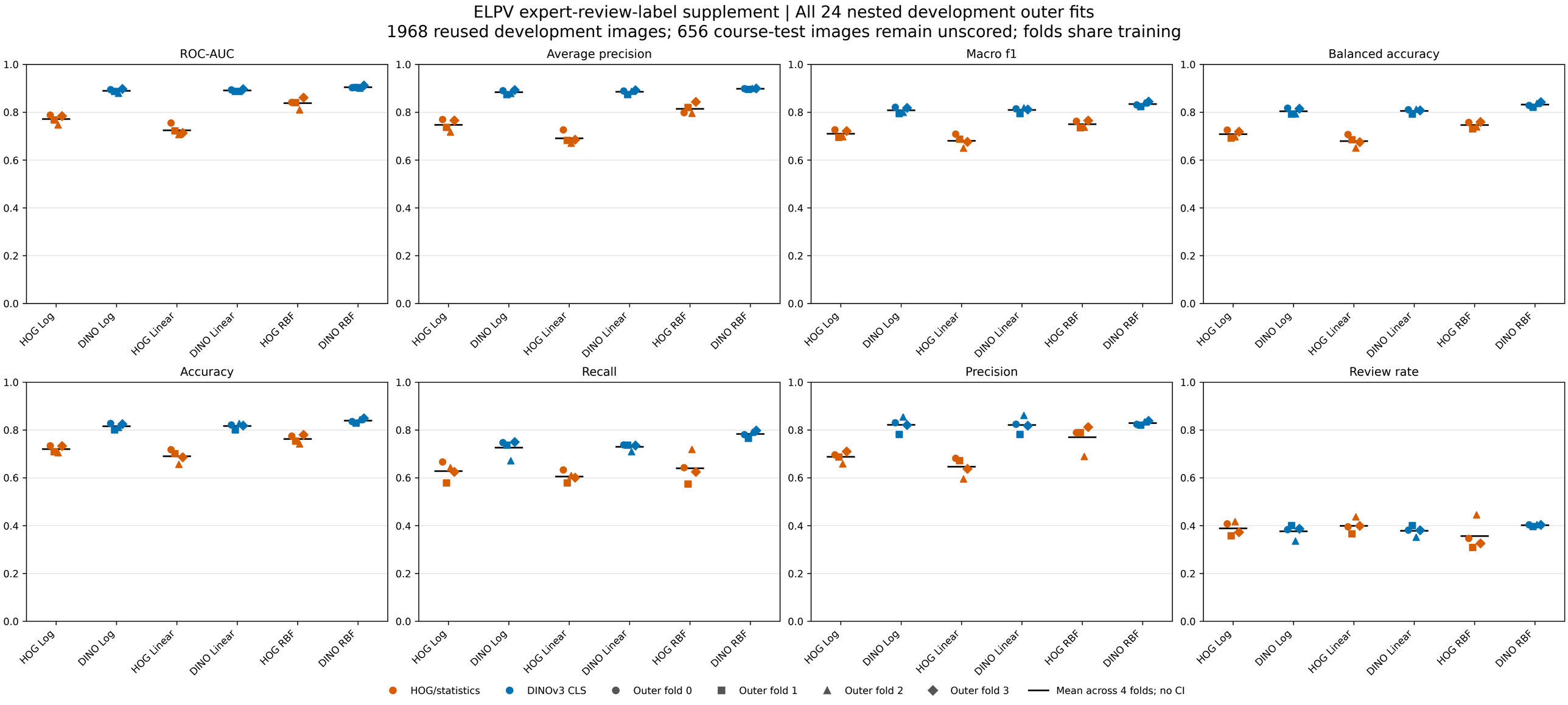
Fitted seed 11 Fitted seed 22 Fitted seed 33 | Mean of fitted-seed points; no CI


Representation and reference mechanisms | End-to-end source inference latency
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
dinov3_patch_mlp
dinov3_postlora_no_reference
dinov3_postlora_reference
dinov3_global_normal_distance
dinov3_postlora_reference_b20
dinov3_global_normal_distance_b20
1.0
10.0
100.0
Pipeline ms/image; median of 3 passes (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


Representation and reference mechanisms | Observed RSS highwater
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
dinov3_patch_mlp
dinov3_postlora_no_reference
dinov3_postlora_reference
dinov3_global_normal_distance
dinov3_postlora_reference_b20
dinov3_global_normal_distance_b20
256
512
1024
2048
4096
Observed process RSS highwater, MiB (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


Full published-method ports | End-to-end source inference latency
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
supervised_b20
supervised_b100
ssl_primary
ssl_ensemble
1.0
10.0
100.0
Pipeline ms/image; median of 3 passes (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


Full published-method ports | Observed RSS highwater
M2 Max 32 GiB / MPS; all registered systems and fitted seeds retained
SCDD source inputs (351 images); batch 1
SCDD source inputs (351 images); batch 8
UCF source inputs (430 images); batch 1
UCF source inputs (430 images); batch 8
supervised_b20
supervised_b100
ssl_primary
ssl_ensemble
256
512
1024
2048
4096
Observed process RSS highwater, MiB (log scale)
Fitted seed 11
Fitted seed 22
Fitted seed 33
Mean of fitted-seed points; no CI


ELPV expert-review-label supplement | All 24 nested development outer fits
1968 reused development images; 656 course-test images remain unscored; folds share training
ROC-AUC
Average precision
Macro f1
Balanced accuracy
Accuracy
Recall
Precision
Review rate
1.0
0.8
0.6
0.4
0.2
0.0
HOG Log
DINO Log
HOG Linear
DINO Linear
HOG RBF
DINO RBF
HOG/statistics
DINOv3 CLS
Outer fold 0
Outer fold 1
Outer fold 2
Outer fold 3
Mean across 4 folds; no CI